KS-FIM Investigation Report: Methodology, Reasoning, and Analysis

Repo: `hari251086/KS-FIM` | **Author:** Harishkumar Sellamuthu **Investigation period:** 2026-08-05 (single session, Phases 1-6) **Status:** Core question closed and falsified (exact, reproducible). Extension question (issue #5) left open, inconclusive.

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1. Executive Summary

A 2021 COSPAR presentation ("Regularized Orbit Observability for Resident Space Objects," PEDAS.1-0023-21) claimed that satellite-tracking observability — measured as the determinant of a Fisher Information Matrix (FIM) built from range measurements — is quantitatively "stronger" when the FIM is computed in Kustaanheimo-Stiefel (KS) regularized coordinates than in Cartesian coordinates, for the same physical tracking scenario.

This investigation built an independent, from-scratch implementation of both FIMs against KSROP's own tested KS-transformation code, and tested the claim empirically and analytically across six escalating phases. The result:

- **The claim, as the source measured it (raw 4×4 vs. 3×3 determinant comparison), is false.** The 4×4 KS-space FIM is structurally rank-deficient (rank ≤ 3) in every tested case — the "advantage" the presentation reports is not new physical information, it is an unnormalized units/dimension artifact.
- **The artifact has an exact closed form.** Once correctly rank-matched, the KS/Cartesian information ratio equals exactly `64·r³` (`r` = the satellite's instantaneous orbital radius), independent of station geometry, count, or measurement noise. This was verified to 10^{-8}–10^{-11} relative precision across dozens of orbits and a full 144-point, 2-revolution time series.
- **The artifact extends to conditioning, not just magnitude.** The FIM's condition number — a dimensionless, rank-matched metric relevant to numerical filter behavior — is *also* exactly representation-invariant.
- **A deeper, genuinely different question remains open.** Whether actual *sequential nonlinear filtering* (an EKF tracking a satellite over time, not a single-instant FIM snapshot) shows any representation- dependent effect was tested directly with a full covariance Kalman filter. The result is **inconclusive** — a numerically jagged divergence pattern, most consistent with finite-difference numerical noise rather than a real physical effect. This is reported honestly as unresolved, not forced into a false conclusion either way.

92/92 automated tests pass. All code, tests, and documentation are committed to `HS-dev` and `main` (fast-forward merged after every phase).

2. Origin and Motivating Claim

The source material is a COSPAR conference presentation reporting FIM observability metrics for four real HEO objects (35497, 37151, 39615, 42928) and one GEO object (28868), tracked by named ground stations (Kwajalein, Ascension, SHAR, Tenerife, and others for the GEO case). Its core empirical claim: the raw determinant of a 4×4 KS-space FIM is routinely 10^4–10^{18} times larger than the corresponding 3×3 Cartesian FIM's determinant for the identical tracking geometry, interpreted as KS coordinates providing genuinely superior orbit-determination observability.

Two structural red flags motivated building this repo instead of accepting the claim at face value:

1. **Dimensional mismatch.** Physical position has exactly 3 true degrees of freedom. The KS map `r = (x, y, z, 0) = L(u)u` embeds this 3-DOF position into a 4-dimensional `u`-space via the Hopf fibration — the 4th dimension is a well-known redundant "gauge" direction with no independent physical content. A range measurement `h(r) = |r − R_station|` depends only on physical position, so a FIM built from such measurements, however it's parameterized, can never carry information about a direction the measurement doesn't see. A 4×4 matrix built this way should therefore be structurally rank-deficient (rank ≤ 3), not full rank as raw nonzero 4×4 determinants would imply. 2. **Unnormalized magnitude comparison.** Comparing a raw 4×4 determinant (units: km^8/σ^8, roughly) against a raw 3×3 determinant (km^6/σ^6) directly is not a like-for-like comparison — it mixes physical units across a coordinate change with its own Jacobian scale factor. A magnitude difference under such a comparison is the *expected* signature of a units artifact, not evidence of new information.

Both suspicions turned out to be correct and became the spine of Phases 1 and 4 respectively.

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3. Methodology

3.1 General approach

The guiding principle throughout: **never take either the source presentation's numbers or this repo's own derivations on faith — build independent ground truth and check everything against it, twice where possible.**

- **Ground truth for the KS transformation**: KSROP's own `car2ks` / `ks2car` routines (`src/Subrouts.F`), already independently tested across hundreds of cases in that repo, not a re-derivation of the KS map. This repo's `fim.F` builds FIMs *on top of* KSROP's transformation rather than reimplementing it.

- **Two independent gradient derivations.** The KS-space range gradient needed for each FIM was computed two ways that don't share an assumption: (a) central finite difference through the real `ks2car` map (Phase 1, deliberately *not* the source presentation's own analytical `du/dr` chain-rule slide, so this doesn't inherit any error in that derivation), and (b) an independently-derived analytical gradient from the textbook KS differential identity `dr = 2·L(u)·du` (Phase 2). Agreement between the two to ~10^{-10} relative precision (Phase 2) is what let later phases trust the analytical form for performance and cleaner derivations without re-litigating Phase 1.
- **Rank via eigenvalues, never via raw determinant.** A cyclic Jacobi eigenvalue solver (`src/linalg.F`, `jacobi_eig`) was built from scratch (no existing eigensolver anywhere in KSROP) specifically because a raw determinant can be a small but nonzero float even when an eigenvalue is genuinely zero to numerical precision — the actual rank-deficiency question is an eigenvalue question. All comparisons in this repo use a "reduced determinant" (product of only the eigenvalues above a `1e-8`-relative threshold) rather than the raw determinant of the full matrix.
- **Rank-matched, not fixed-size, comparisons.** An early implementation bug (caught during Phase 1 test development) hardcoded "always compare the top 3 eigenvalues." With fewer than 3 independent range directions (e.g. 2 stations), the true rank is 2 on *both* sides, and forcing a 3rd eigenvalue into the product corrupted the comparison, including producing nonsense negative ratios. The fix — always use `min(true rank on each side)` — became a standing rule for every subsequent phase.
- **Real case-study reproduction before synthetic generalization.** Each phase's synthetic test cases (varied orbits, station counts) were paired with reproduction of the source presentation's own named objects and stations wherever the source published enough data to do so (`ksfim_case_study.F`), so findings are anchored to the actual claim being tested, not only to convenient synthetic geometry.

3.2 Toolchain

- Fortran, `fpm` package manager, fixed-form source with implicit typing (`implicit double precision (a-h,o-z)`), matching the KSROP/OREM/KS-Pc house style.
- Compiled and tested with Intel `ifx` 2025.0 on Windows (Visual Studio + Intel oneAPI environment variables required in the same shell session).
- KSROP consumed as a real `fpm` git dependency (tag `v2.2.0`), not hand-copied files.

3.3 Escalation structure

Each phase was scoped to *strengthen or extend* the prior phase's finding rather than simply add more of the same, and each phase's own results were used to sharpen the next question:

Phase	Question	Method	Outcome
1	Is the 4×4 KS FIM really rank-deficient, and is the ratio a units artifact?	Finite-difference gradient, from-scratch eigensolver	Yes to both — falsified
2	Is Phase 1 a finite-differencing artifact? Does real station geometry (GMST) change anything?	Independent analytical gradient; real per-epoch station rotation	No and no — strengthened
3	Does the finding hold at the extreme edge (GEO, 1 station)?	Extended case study + dedicated edge test	Yes, generalizes a caveat the source itself only partially noted
4	Is the "constant ratio" actually a *predictable* number?	Analytical derivation from KS conformal-map identities	Yes — exact closed form `64r³`, verified analytically and empirically
5	Is there *any* correctly-normalized comparison where KS wins?	Condition number (dimensionless, rank-matched)	No — also exactly invariant, a direct corollary of Phase 4
6	Does *sequential nonlinear filtering* (not a single snapshot) show an effect the FIM can't see?	Full covariance EKF over a real multi-step timeline	**Inconclusive** — first non-clean result, reported as such

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4. Phase-by-Phase Analysis

4.1 Phase 1 — Core Finding: Rank Deficiency and the Units Artifact

Built: `src/linalg.F` (Jacobi eigensolver), `src/fim.F` (`fim_build_cartesian`, `fim_build_ks` via finite-difference gradient, `fim_reduced_det_ks` / `fim_reduced_det_cartesian`), `app/ksfim_case_study.F` (reproduces the source's 4 real HEO objects with their published elements and station table).

Findings:

- `rank(F_ks) = min(nstat, 3)` exactly, in every tested case (3 orbital regimes × 4 station counts synthetically, plus all 4 real case-study objects) — never rank 4. Where a 4th eigenvalue exists numerically it is 8–18 orders of magnitude smaller than the largest, consistent with floating-point noise around a true zero.
- For a *fixed* satellite position, the rank-corrected ratio `reduced_det_ks / reduced_det_cartesian` across 5 wildly different station configurations (different counts, geometries, σ) was `3.7257368×10¹⁴` in *every* case, agreeing to 7–8 significant figures — as clean a confirmation as an empirical test can produce that the "advantage" is a predictable,

position-dependent Jacobian/conformal- scaling artifact (`L(u)ᵀL(u) = |u|²I₄`), not new physical information.

- Across the 4 different real case-study objects, the corrected ratio *did* vary (9.6×10^{13} to 2.4×10^{13}) — consistent with depending on each object's own position, not universal across objects, which is the correct signature (constant only for a *fixed* position across *different* station geometries, the actual claim under test).

Verdict at this stage: not a rejection of KS regularization itself — its propagator-accuracy benefits are real, well-established, and unrelated to this finding — but a specific falsification of "KS gives better FIM-based observability" as the source measured it.

4.2 Phase 2 — Independent Analytical Cross-Check + Real Station Geometry

Built: `src/timeconv.F` (`gmst_deg` , IAU-1982 formula, verified against the published J2000.0 reference value to `1e-4` deg; `geodetic_to_eci`), `range_grad_ks_analytical` / `fim_build_ks_analytical` (`src/fim.F` , via the textbook `dr = 2·L(u)·du` differential identity).

Reasoning: Phase 1's gradient was computed by finite-differencing `ks2car` . Two questions this doesn't answer on its own: (a) is the result a numerical-differencing artifact, and (b) does the earlier fixed-offset station placement (not real per-object GMST rotation) matter?

Findings:

- The analytical gradient reproduces the finite-difference reduced determinant to full displayed precision (~10^{-10} relative, identical digits) for both synthetic cases and all 4 real objects — Phase 1's findings are not a numerical-differencing artifact.
- Adding real per-object epoch/GMST station placement left the **rank-corrected** ratio bit-for-bit *unchanged* for every object, exactly as predicted (it depends only on satellite position). The **raw** ratio, by contrast, changed substantially under the identical geometry change — object 42928's raw ratio flipped sign entirely (`+1.50×10³ → -2.05×10⁴`). A negative "observability" is physically nonsensical for any legitimate information metric — a second, independent line of evidence (beyond the magnitude argument) that the raw comparison is numerically unsound, not just unnormalized.

41/41 tests passing.

4.3 Phase 3 — GEO Case and the Low-Station-Count Edge

Built: extended `ksfim_case_study.F` with the source's GEO object (28868, ANIK F1R — no orbital elements published for this case in the source, so a representative near-GEO state is used, documented in the app's header) and its 1-/2-station sub-cases (Kermit; Kermit+Fairbanks); `test_fim_geo_edge.F` .

Findings:

- `rank(F_ks) = rank(F_cartesian) = nstat` holds exactly down to a *single* observing station (rank 1) — the raw 4×4 determinant is numerically zero (~10^{-57} to ~10^{-27}) at both 1 and 2 stations.
- This generalizes a caveat the source presentation itself only noted for near-degenerate *close station pairs* ("if two stations are very near, observability can go to zero, misconstrued as a single station") into a structural fact: the raw metric collapses to zero at *any* low station count, not just near-coincident geometry.
- A units bug was caught and fixed here: `ksfim_case_study.F`'s station σ had been mistakenly set to the station *altitude* column values (0.01–2.39 km) instead of the source's actual published `σ = 10 km`, present since Phase 1. Confirmed magnitude-only (σ enters both FIMs identically per station, so it cannot change rank or the rank-corrected ratio) — fixed anyway for numerical fidelity to the published numbers.

45/45 tests passing.

4.4 Phase 4 — The Exact Closed Form

Reasoning: Phase 1 established the ratio is *constant* across station geometry for a fixed position. That raises an obvious next question: is the constant itself *predictable*, or just empirically stable? This phase derives it analytically first, then verifies.

Derivation: `L(u)` satisfies the standard KS identity `L(u)ᵀL(u) = |u|²I₄` — `L` is a uniformly-scaled orthogonal (conformal) map at any fixed `u`. Its 3-row physical submatrix `L₃(u)` therefore satisfies `L₃(u)L₃(u)ᵀ = |u|²I₃`, meaning all 3 singular values of `L₃(u)` equal `|u|` exactly. Since `F_ks = JᵀF_cartesianJ` with `J = ∂r/∂u = 2·L₃(u)` (the Phase 2 differential identity), `F_ks`'s nonzero eigenvalues equal exactly `4|u|² ·` the corresponding `F_cartesian` eigenvalues. Combined with the KS regularizing identity `|u|² = r`:

```
reduced_det_ks / det_cartesian = (4|u|²)³ = 64·r³
```

— exactly, independent of station geometry, count, or σ; a single number determined entirely by the satellite's instantaneous distance from the primary.

Built: `test_fim_ratio_formula.F` (10 varied orbits × station geometries), `app/ksfim_timeseries.F` (object 35497 tracked across 2 full revolutions, 144 samples, mean-anomaly-stepped via `oe2car`).

Findings:

- `test_fim_ratio_formula`: 20/20 passing, reldiff `~1e-8` – `1e-10` in every case, including a direct check of the `|u|² = r` identity itself.
- `ksfim_timeseries`: the closed-form prediction held at *every one* of 144 sampled points across 2 full revolutions (`r` ranging 6635.62– 26894.78 km, perigee to apogee) — max relative deviation `1.474×10⁻⁹`. This directly explains why the source presentation's own plots show

observability "evolving" over a revolution: it is fully explained by `r` 's own variation along the orbit, not an independent KS-space time-dependent effect.

65/65 tests passing.

4.5 Phase 5 — Condition Number Invariance

Reasoning: Issue #5 (filed during initial scoping) asked whether *any* correctly-normalized, rank-matched comparison could reveal a real KS advantage. Phase 4's result already implies an answer for the next natural dimensionless candidate: since *every* nonzero `F_ks` eigenvalue scales by the *same* factor (`4|u|²`) relative to its `F_cartesian` counterpart, their *ratios to each other* — the condition number, `λ_max/λ_min` among the rank-matched eigenvalues — must be exactly invariant too, not just approximately.

Built: `test_fim_condition_number.F` (same 10 orbits/station geometries as `test_fim_ratio_formula`).

Findings: `condition_number(F_ks) = condition_number(F_cartesian)` exactly, confirmed to `~1e-8` – `1e-11` relative precision (limited only by eigensolver precision) across all 10 cases. KS coordinates offer no conditioning advantage in the linear/FIM sense either — closing off essentially every reading of "is there a real KS advantage" that a single-instant FIM-based comparison could support.

75/75 tests passing.

4.6 Phase 6 — Sequential EKF Study (Inconclusive)

Reasoning: Phases 1–5 exhaustively address the *single-instant, linear FIM* question. A genuinely different question remained: does actual sequential nonlinear filtering — an EKF tracking a satellite over a real multi-step timeline, with repeated re-linearization — show a representation-dependent effect that a snapshot analysis structurally cannot see? This was explicitly scoped by the user as worth a real investment ("Full EKF simulation study") rather than deferred.

Design principles:

- **Isolate linearization behavior from propagator accuracy.** KS's propagator-accuracy advantage (smoother numerical integration near periapsis, etc.) is real, well-established, and *not* what's under test — conflating the two would make any result uninterpretable. The reference trajectory therefore uses an **exact** two-body Kepler flow map (`kepler_flow_cartesian` : `car2oe` → mean-anomaly advance → `oe2car`), never numerically integrated, so any effect found is attributable purely to linearization/filtering, not propagation error.
- **Reuse validated building blocks wherever possible.** The measurement update reuses the exact `range_grad_cartesian` / `range_grad_ks_analytical` rows already validated in Phases 1–4. State-transition matrices are computed by central finite difference of the exact flow

map (`stm_cartesian` , `stm_ks`), following the same finite-differencing philosophy established in Phase 1 for `range_grad_ks` , rather than a newly hand-derived Jacobian.

- **A physically fair, matched initial condition.** `P0_cart` is a diffuse 1 km/1 m/s prior; `P0_ks` is *not* an independently chosen prior but `P0_cart` mapped through `car2ks_jacobian` (a finite- difference Jacobian of `car2ks` itself) — an honest test needs the same physical uncertainty represented in both coordinate systems, not two separately-guessed starting points that could bias the comparison either way.
- **Standard covariance Kalman recursion**, `Q = 0` (no artificial process noise), so the comparison isolates pure linearized-uncertainty transport: predict `P ← ΦPΦᵀ` ; update via sequential scalar measurements (`src/kalman.F` , one station at a time — mathematically equivalent to a batch update for independent measurements, but needs only a scalar division, never an N×N inverse).

Bug found #1 (real, previously latent): `w = dsqrt(-0.5d0(amue/ a_kep(1)))` *, used in* every *prior phase (11 occurrences across 10 files), computes* `sqrt` *of a negative number —* `NaN` *. The correct formula, matching KSROP's own* `driver_KS.F` / `test_subrouts.F` *convention, is* `w = sqrt(0.25·μ/a) = sqrt(-E/2)` *(both the sign* and *the coefficient were wrong). This never affected any Phase 1–5 finding:* `w` *only enters the KS* velocity *(* `us` *) via* `ks2car` / `car2ks` *; every prior computation depends only on* position* (`u` , via `ks2car` 's `x` output, which is `w` -independent by construction). Phase 6 is the first computation in this repo to actually use KS velocity — which is what exposed a bug that had been silently producing `NaN` in an unused variable for five phases. Fixed repo-wide; the full 92-test suite (up from 75) still passes 100% after the fix, empirically confirming the bug was genuinely inert everywhere it had previously appeared.

Bug found #2 (metric design flaw, caught before publishing a result): An initial naive comparison of the *full* 6×6 (Cartesian) vs. 8×8 (KS) state covariance's condition number produced an enormous, implausible divergence (up to `8.4×10⁵` × relative difference). Diagnosis: this comparison conflates position with velocity (different physical units in each representation — km/s vs. the KS `us` scale) and ignores that KS's `u` -block carries a structural gauge redundancy Cartesian's `r` -block does not — the two full-state matrices are not dimensionally the same kind of object even when representing identical physical uncertainty. This is the same category of mistake Phase 1 originally diagnosed in the *source presentation's* raw-determinant comparison, now recurring in this repo's own draft methodology. **Fix**: Schur-complement position marginalization — `P_pos = P_pp − P_pv·P_vv⁻¹·P_vp` — computed via a new `mat_inverse` (Gauss-Jordan with partial pivoting, `src/linalg.F`), yielding a position-only covariance in each representation (3×3 Cartesian, 4×4 KS, rank-matched via the same eigenvalue-threshold logic used throughout the repo). This is the direct sequential analog of Phase 5's already-proven, dimensionally-fair metric.

Result, even after both fixes: **inconclusive.**

- `rank(P_pos_ks) ≠ rank(P_pos_cart)` at 71 of 143 steps (KS drops to rank 2 while Cartesian holds steady at rank 3).

- The condition-number ratio ranges from `~0.1×` to `~7.9×10⁴×` across the run — not converging toward a constant the way every Phase 1–5 result did.
- **Decisive diagnostic**: the divergence pattern is *jagged*, not smooth. `cond_cart` varies gently step to step, tracking the smoothly- varying orbital geometry, while `cond_ks` spikes by 2–3 orders of magnitude between *adjacent* steps (e.g. steps 57→58→59: `1.6×10³ → 5.0×10⁷ → 1.9×10³`). A real physical effect tracking a smooth orbital trajectory should itself vary smoothly; this pattern is instead the classic signature of numerical sensitivity compounding through 144 sequential finite-difference-STM and Schur-complement- inversion operations.

Disposition: this result does **not** confirm a representation- dependent nonlinear-filtering effect, and does **not** confirm its absence either — unlike Phase 5's exact, reproducible-to-10^{-11} result, the finite-difference-STM approach used here is not precise enough to distinguish a genuine effect from numerical noise at the observed scale. A decisive answer would need an **analytically-derived** (not finite- differenced) KS state transition matrix — a materially larger derivation effort, left as explicitly unstarted future work rather than attempted under this session's scope. This is reported honestly as inconclusive, consistent with the standing principle applied throughout this investigation: only publish findings that are clean and reproducible.

92/92 tests passing (12 new: `test_dynamics` — exact-flow round-trip, energy conservation, and the `2·L₃(u)·(∂u/∂r) = I₃` right-inverse identity; `test_kalman` — scalar update vs. a hand-computable 1D case; `test_mat_inverse` — Gauss-Jordan vs. `A·A⁻¹ = I`).

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5. Bugs Found and Fixed — Full Ledger

A chronological record of every real defect caught during this investigation, in the order found:

#	Phase	Defect	Root cause	Impact	Fix
1	1	Rank-mismatch in reduced-determinant comparison	Hardcoded "always top-3 eigenvalues" regardless of true rank	With <3 stations, corrupted comparisons, including negative ratios	Use `min(true rank)` on each side
2	3	Station σ set to altitude values (0.01–2.39 km) instead of published `σ=10 km`	Copy/reuse error from the settings box shared with the HEO cases	Magnitude-only (σ enters both FIMs identically)	Corrected to published value
3	6	`w = sqrt(-0.5·μ/a)` computes `NaN` — wrong sign *and* coefficient	Formula never cross-checked against KSROP's own convention; silently inert since nothing used KS velocity	None on Phases 1–5 (position-only); would have corrupted any future KS-velocity use	Fixed repo-wide to `sqrt(0.25·μ/a)`, all 11 occurrences
4	6	Full 6×6/8×8 state condition-number comparison isn't dimensionally fair	Conflates position/velocity units, ignores KS's gauge redundancy	Produced a spurious `~10⁵` ×–`10⁸` × "finding" before the fix	Schur-complement position marginalization

Notably, bugs #1 and #4 are the *same category of error* — an unnormalized or rank-mismatched comparison producing an artificially inflated "difference" — recurring first in the source presentation's own methodology (which this repo exists to critique) and then, independently, in this repo's own draft Phase 6 metric. Catching #4 before publishing a result is itself evidence the investigation's own standards were applied consistently, not just to the object of study.

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6. Key Mathematical Results

Rank-deficiency (Phase 1, proven analytically in `ALGORITHM.md` §5): since `F_ks = JᵀF_cartesianJ` with `J = ∂r/∂u` (3×4, because physical position `r` has only 3 independent components), the full KS-space FIM can never exceed `rank(F_cartesian) ≤ 3`, regardless of station count or geometry — a structural property of the transformation, not an empirical coincidence.

Exact closed-form ratio (Phase 4):

```
reduced_det_ks / det_cartesian = 64·r³
```

where `r` is the satellite's instantaneous orbital radius. Derived from `L(u)ᵀL(u) = |u|²I₄` (KS conformality) and the regularizing identity `|u|² = r`.

Condition-number invariance (Phase 5, a direct corollary of the above):

```
condition_number(F_ks) = condition_number(F_cartesian)   [exactly]
```

because every nonzero `F_ks` eigenvalue scales by the identical factor `4|u|²` relative to its `F_cartesian` counterpart, so their ratios to each other cancel that factor.

Right-inverse identity (Phase 6, used to validate `car2ks_jacobian`): since `r(u(r)) ≡ r` identically along whatever specific branch `car2ks`'s algebraic construction picks of the KS map's gauge circle,

```
2·L₃(u)·(∂u/∂r) = I₃   [exactly, regardless of gauge branch]
```

— note this is *not* the same as matching the Moore-Penrose pseudo- inverse `L₃(u)ᵀ/(2|u|²)`, which assumes a particular (minimum-norm) branch that `car2ks`'s actual construction has no reason to follow; an early draft of this test incorrectly assumed the pseudo-inverse form and had to be corrected (see `test_dynamics.F` for the full reasoning).

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7. Test Coverage Summary

92/92 passing as of the final commit (`c5107e6`):

Test	Count	Phase	What it checks
`test_fim_cartesian`	3	1	Cartesian FIM vs. hand-computable orthogonal-LOS geometry
`test_fim_ks_rank`	24	1	`rank(F_ks)=min(nstat,3)` across 3 orbital regimes × 4 station counts
`test_fim_reduced_consistency`	1	1	Rank-corrected ratio constant across 5 station configurations
`test_fim_analytical_vs_fd`	9	2	Analytical vs. finite-difference gradient agreement
`test_timeconv`	4	2	GMST vs. published J2000.0 reference; geodetic-to-ECI sanity
`test_fim_geo_edge`	4	3	Rank tracking down to 1-station edge case
`test_fim_ratio_formula`	20	4	Exact `64r³` closed form across 10 orbits/geometries
`test_fim_condition_number`	10	5	Exact condition-number invariance, same 10 cases
`test_dynamics`	12	6	Exact-flow round-trip, energy conservation, right-inverse identity
`test_kalman`	3	6	Scalar Kalman update vs. hand-computable case, trace/symmetry
`test_mat_inverse`	2	6	Gauss-Jordan inverse vs. `A·A⁻¹=I`

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8. Final Verdict

On the original claim: falsified, cleanly and on multiple independent lines of evidence. The COSPAR presentation's raw 4×4-vs-3×3 FIM determinant comparison is not methodologically sound. The 4×4 KS-space FIM is empirically and analytically confirmed rank-deficient (structural rank ≤ 3) in every tested case spanning HEO and GEO regimes, 1–5 station counts, two independent gradient derivations, with and without GMST- accurate geometry, and 10+ varied orbital configurations. Once correctly rank-matched, the resulting KS/Cartesian ratio behaves exactly as a coordinate/Jacobian scaling artifact would: an exact, closed-form function of satellite position alone (`64r³`), constant across station geometry, and with condition number exactly invariant too. This does not mean representation-dependent observability is never a real effect for nonlinear systems in general — a legitimate, different research question — but this specific presentation's evidence for a genuine "KS gives better observability" effect does not survive a rank-correct, same-basis, dimensionally-fair comparison, examined from every angle a single-instant FIM analysis can support.

On the extension question (issue #5): genuinely unresolved. Phase 5 closed the linear-FIM reading definitively (condition number is also exactly invariant). Phase 6 built the actual nonlinear sequential- filtering study needed to test the deeper question directly, but the result was numerically inconclusive rather than a clean confirmation or rejection. This is reported as an open problem with a clear, scoped path to resolution (an analytical KS state transition matrix), not glossed over or forced into a false conclusion.

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9. Open Questions / Future Work

1. **Issue #5 (only remaining open issue)**: does sequential nonlinear filtering show a real representation-dependent effect? Resolving this needs an analytically-derived KS state transition matrix (replacing Phase 6's finite-difference `stm_ks`) precise enough to distinguish a genuine effect from the numerical noise that currently dominates the Phase 6 result. This is a nontrivial derivation (the KS EOM's variational/STM equations), not a quick follow-on — treat as a fresh, explicitly-scoped investigation if picked up again, not "one more phase in the same style" as Phases 1–5. 2. Not investigated and out of scope for this repo: whether *any* correctly-normalized sense of representation-dependent observability exists for nonlinear orbital filtering *in general* (beyond this one presentation's specific claim) — a legitimate, broader research question this repo's falsification does not itself resolve.

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10. Reproducibility

```
git clone https://github.com/hari251086/KS-FIM
cd KS-FIM
fpm build --compiler ifx
fpm test --compiler ifx            # 92/92 passing
fpm run ksfim_case_study --compiler ifx
fpm run ksfim_timeseries --compiler ifx
fpm run ksfim_ekf_study --compiler ifx
```

Requires Intel `ifx` 2025.0 (Visual Studio + Intel oneAPI environment variables in the same shell session on Windows); not yet verified with `gfortran`. Dependency: KSROP (`git+https://github.com/hari251086/KSROP`, tag `v2.2.0`).

Full technical writeup with equations and per-phase validation detail: `ALGORITHM.md`. Project overview and version history: `README.md`. Durable GitHub record of the findings: issues #6 (verdict, updated every phase) and #5 (the still-open extension question).